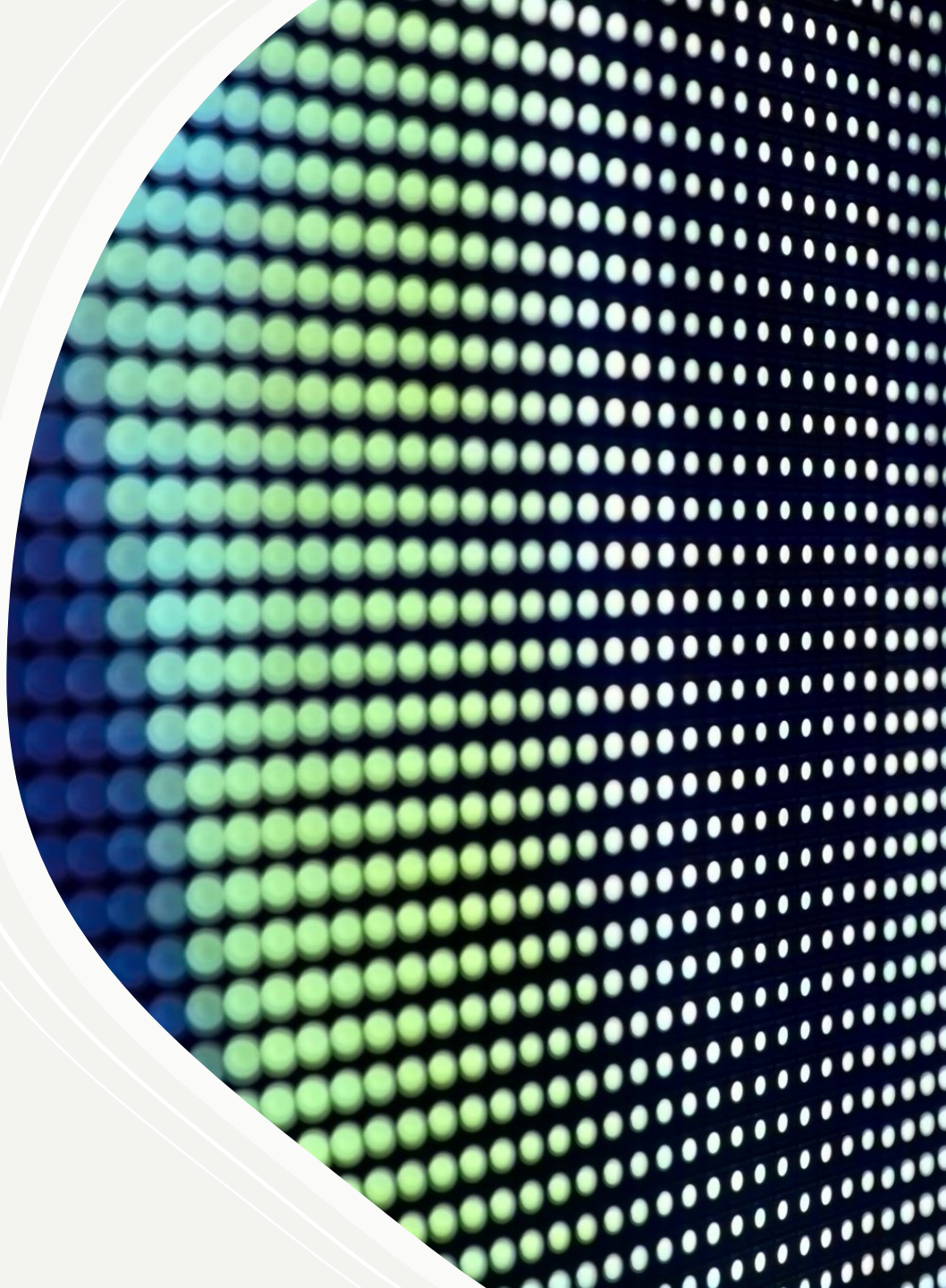
Signal Processing: Konsep, Representasi, dan Operasi

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Outlines

- Building blocks
- Signals: the basics
- Sinusoidal signal
- Sinusoidal signal examples
- Complex Sinusoidal
- Exponential, Step, and Impulse Signals

Basis Expansions to Build Signal with Blocks

Let’s have a look at these images



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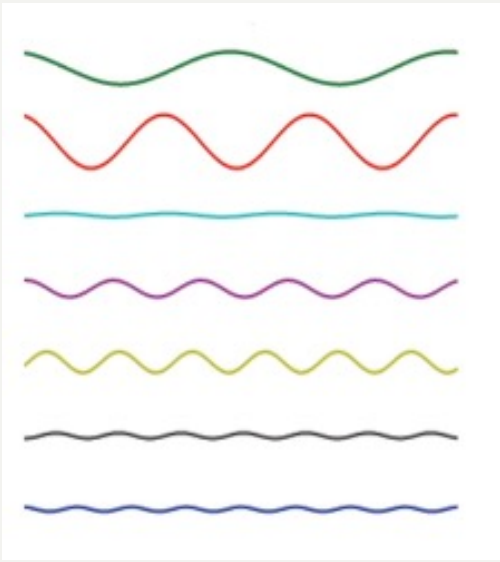
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Complex Signal



Decompose

Signal Dasar



Decomposing complex signals into simple signals

filtering

denoising

compression

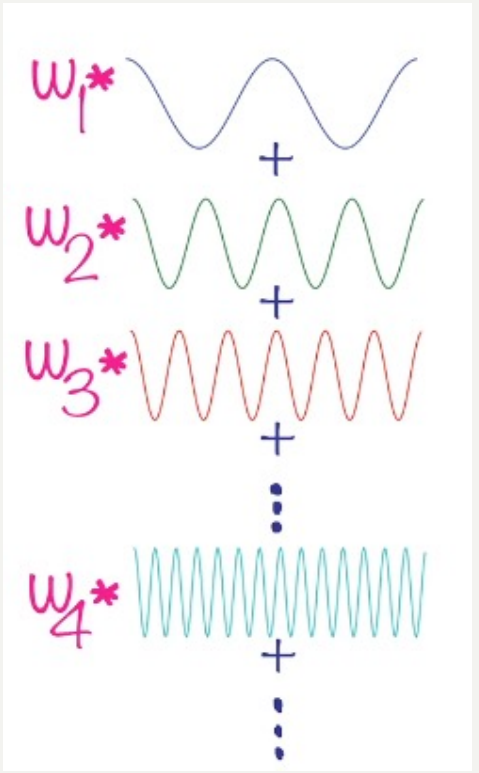
equalisation

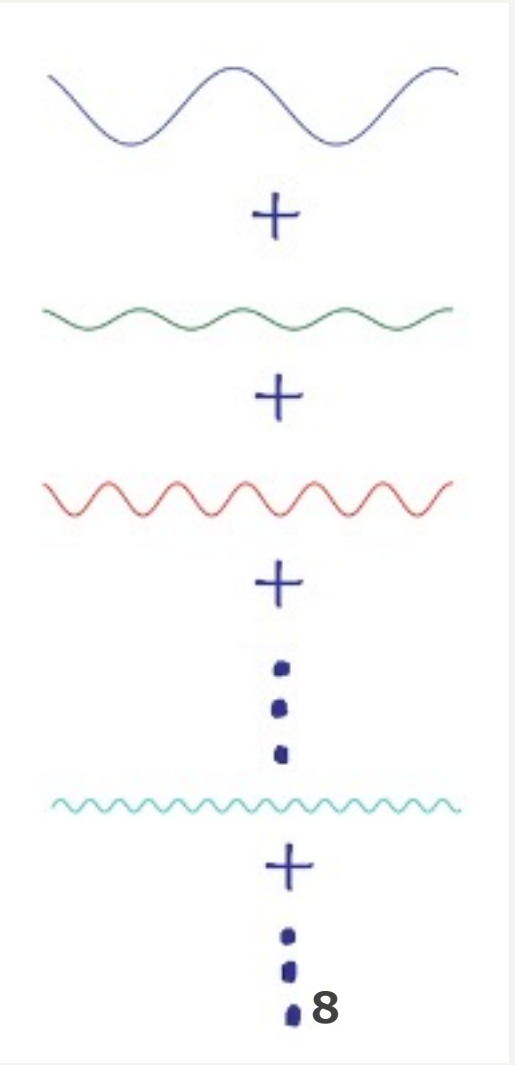
Weighted Sums of Signals







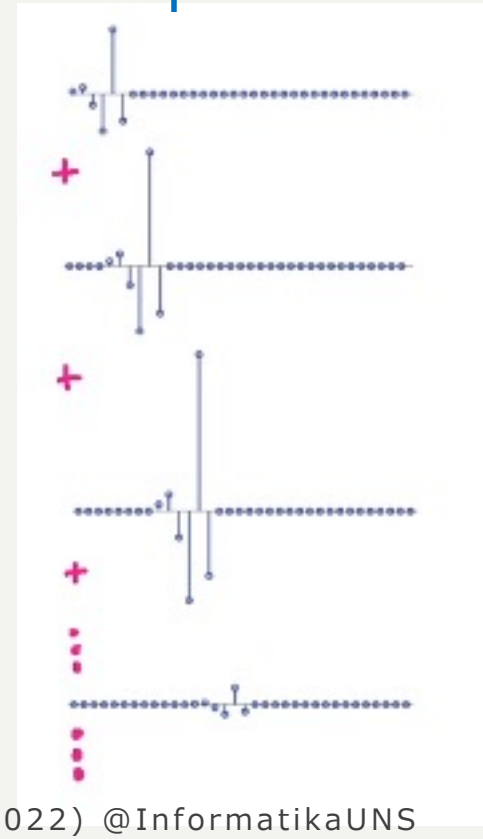

w1*
w2*
w3*
w4*



Signal



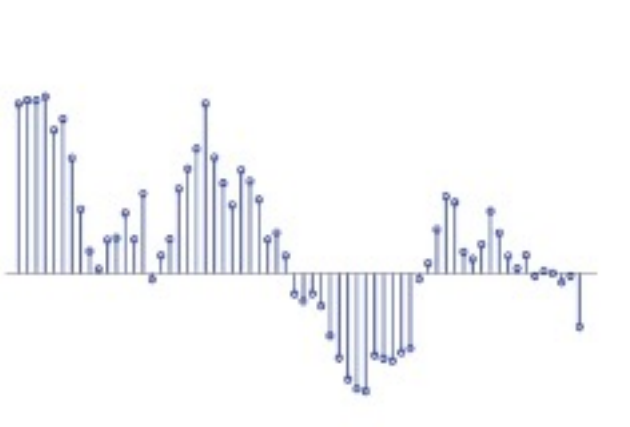
Expansion



Weights

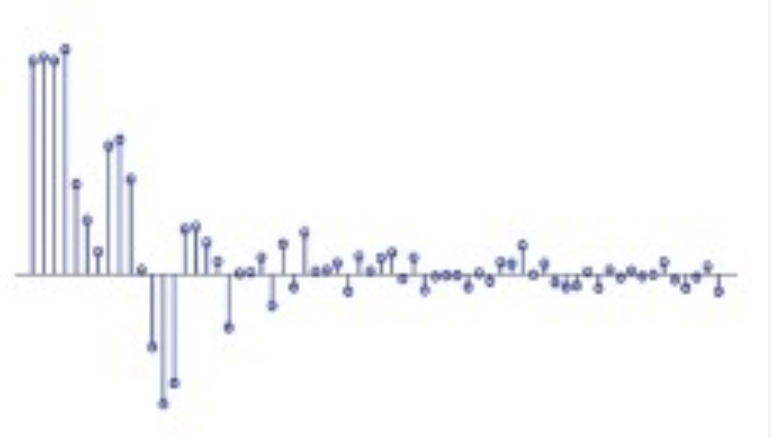


Signal



Equivalent

Weights



$\{x[0], x[1], x[2], \ldots, x[N-1]\}$ $\{w[0], w[1], w[2], \ldots, w[N-1]\}$

Weighted Integral of Bases

$$x(t) = \int x(\Omega)\cos(\Omega t)\,d\Omega$$

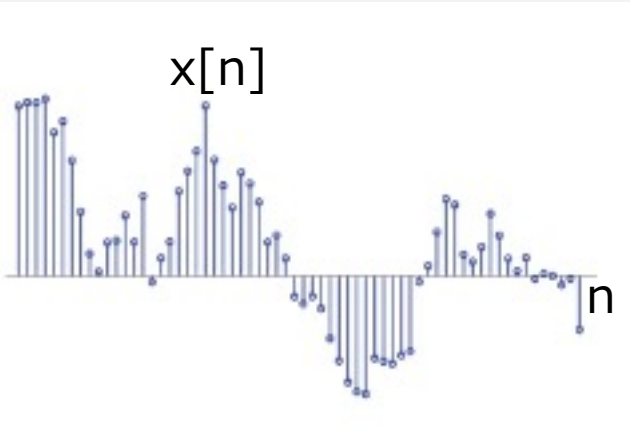
$$= \lim_{\Delta \to 0} \sum_{k} x(\Omega_k)\cos(\Omega_k t)\,\Delta$$

$$\Delta = \Omega_k - \Omega_{k-1}$$

$$\approx \sum_{k} \underbrace{X(\Omega_k)\Delta}_{\text{weights}}\ \underbrace{\cos(\Omega_k t)}_{\text{bases}}$$

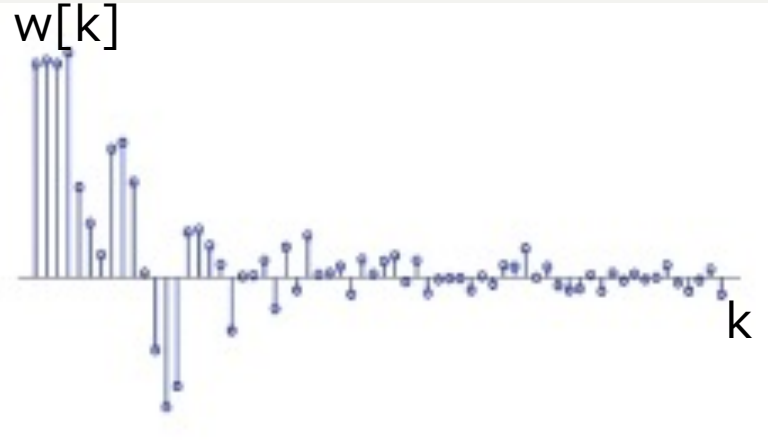
The conversion of Signals into Equivalent set of Weights

Signal



Transformasi

Weights



Discrete fourier transform

Discrete wavelet transform

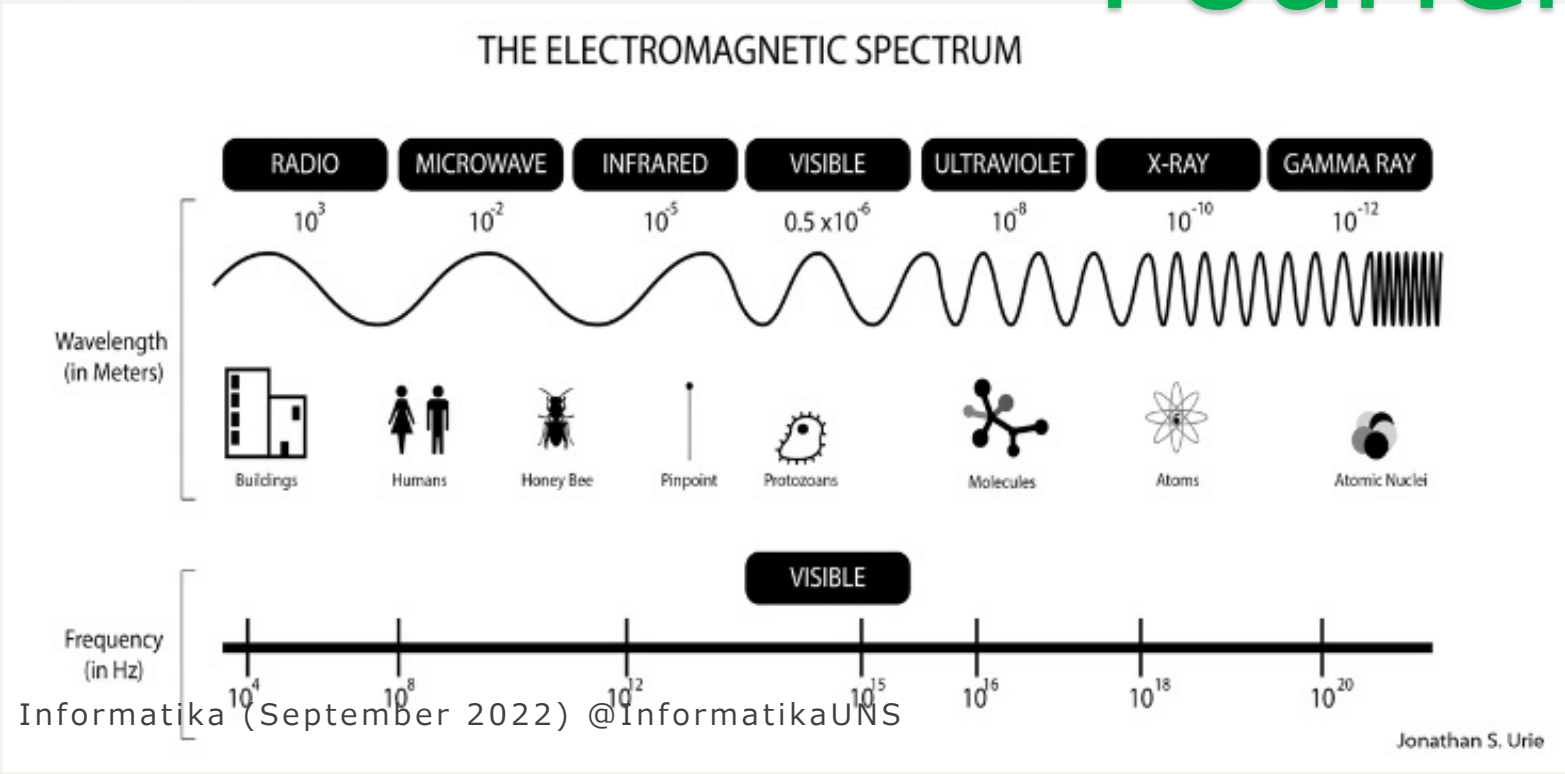
Discrete cosine transform

Choice of basis?

- The choice of of basis for representing arbitrary signals is application dependent. So, **no universally optimal set of basis signals**. Let's say, there are principal components, Haar, Daubachies, Morlet, Biorthogonal, Coiflet, Symlet, Sinusoids, and many more which can be a basis.
- One common class of bases is **sinusoids**. And sinusoids under different arrangement of frequencies are the basis of Fourier transforms and analysis.

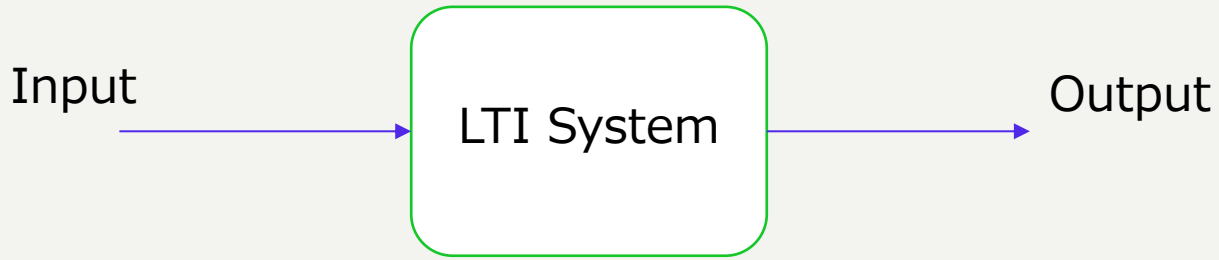
Fourier Transform and Analysis

Sinusoids ⟷ Fourier



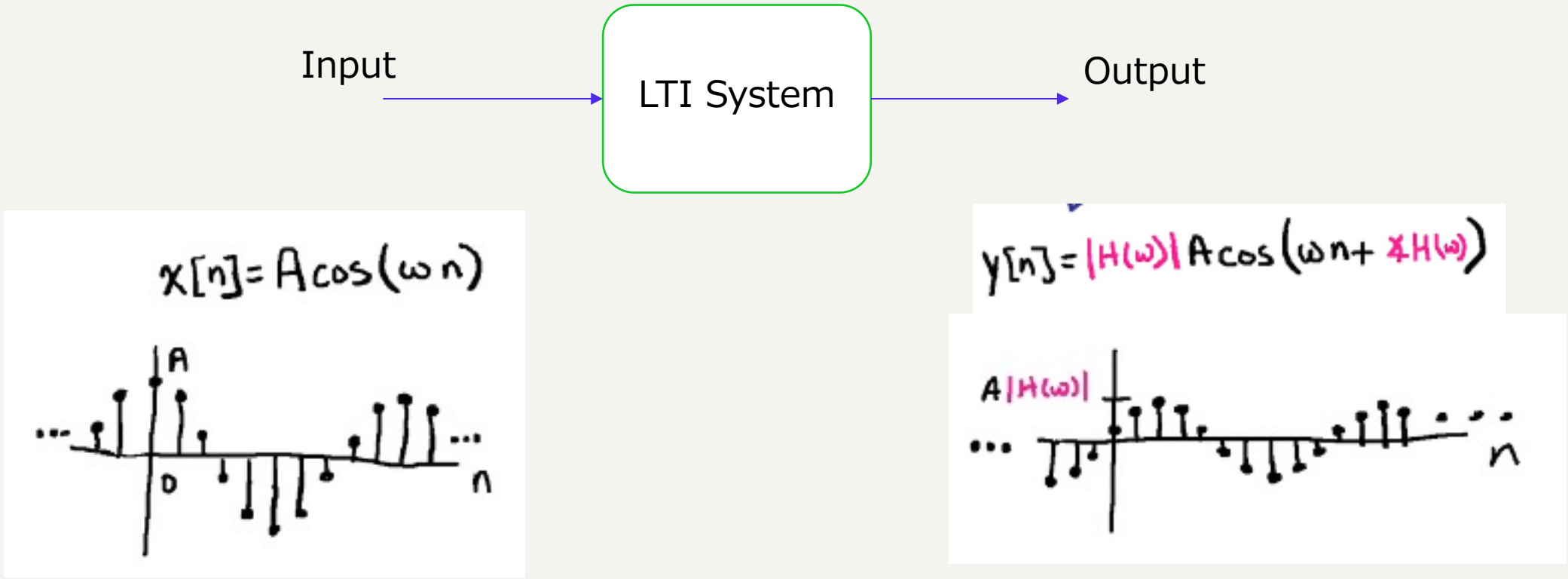
Sinusoid Signals: EigenFunctions

- Sinusoids are “eigenfunctions” of Linear Time Invariant (LTI) Systems and LTI systems are widely used in signal processing.

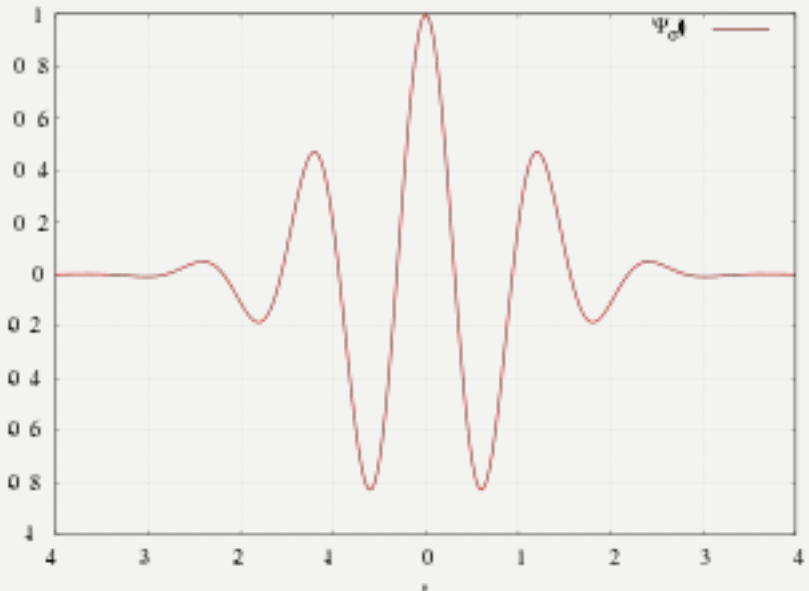


*eigenfunctions → proper or characteristic function

EigenFunctions Properties



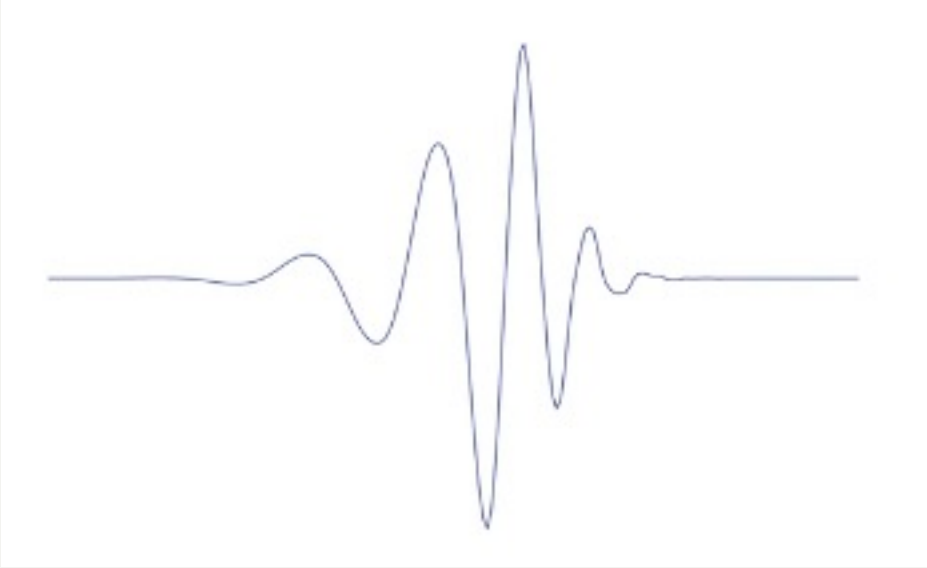
Wavelet Bases



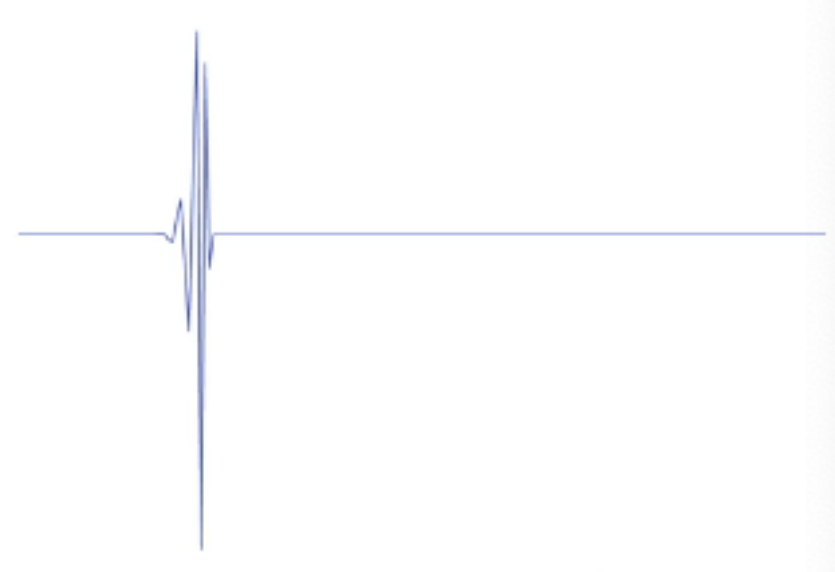
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Duration ↑ ⟷ Oscillation frequency ↓

Time and Frequency Resolution trade-off in Wavelet

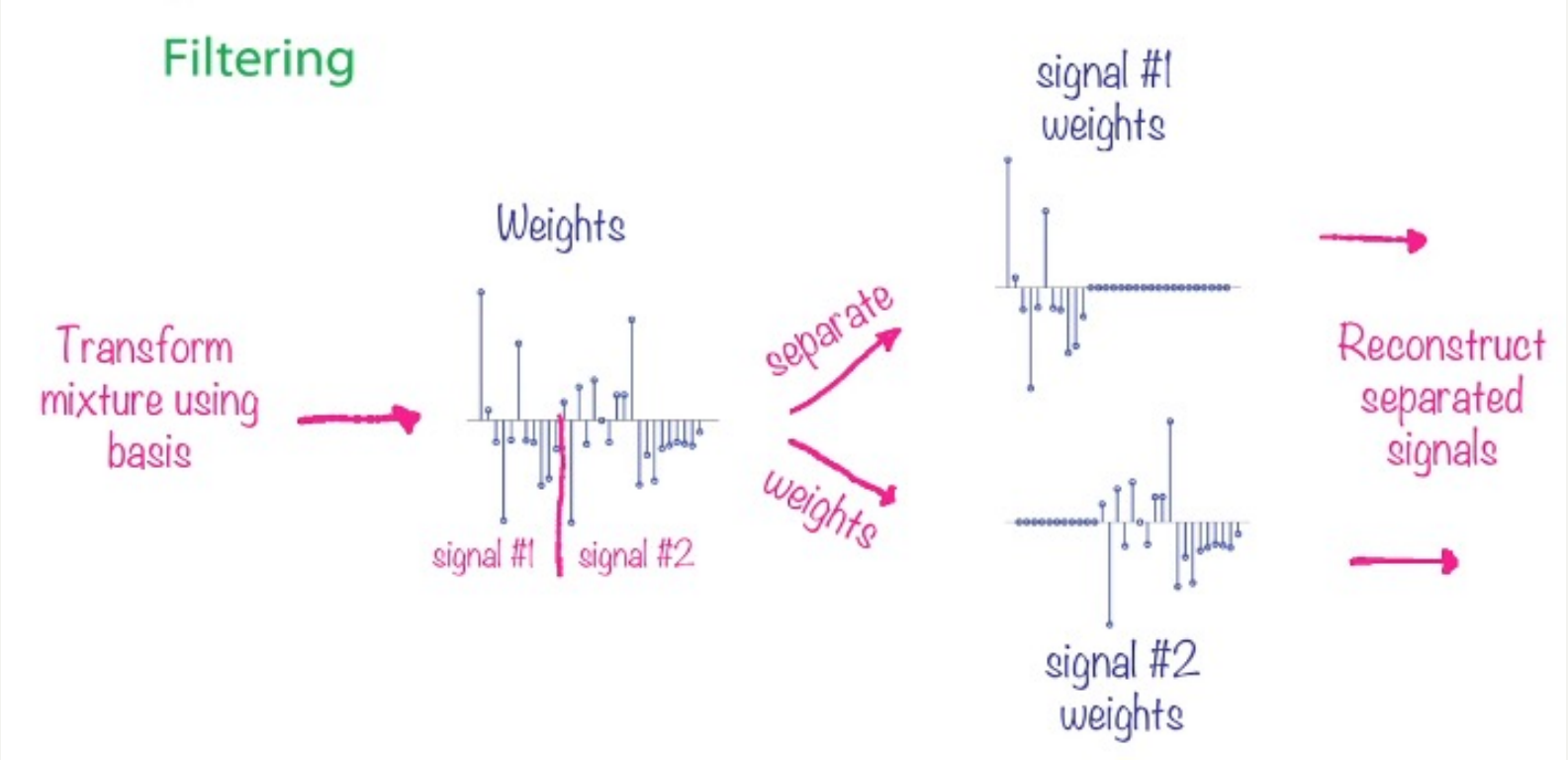


Broad in time
Narrow in frequency

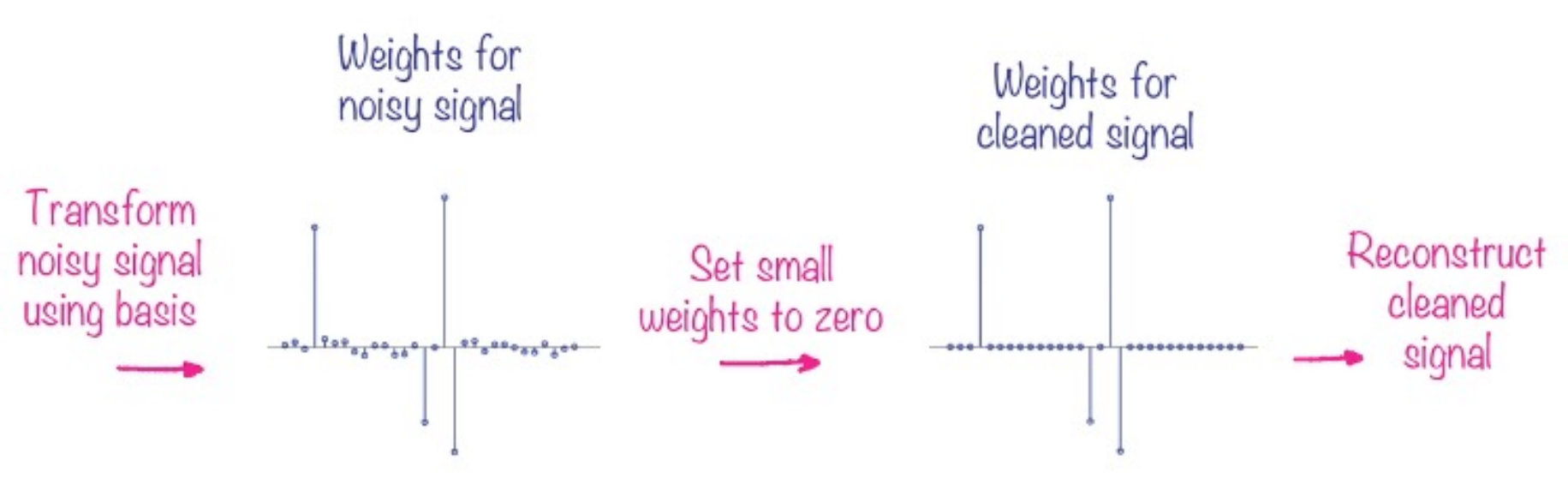


Narrow in time
Broad in frequency

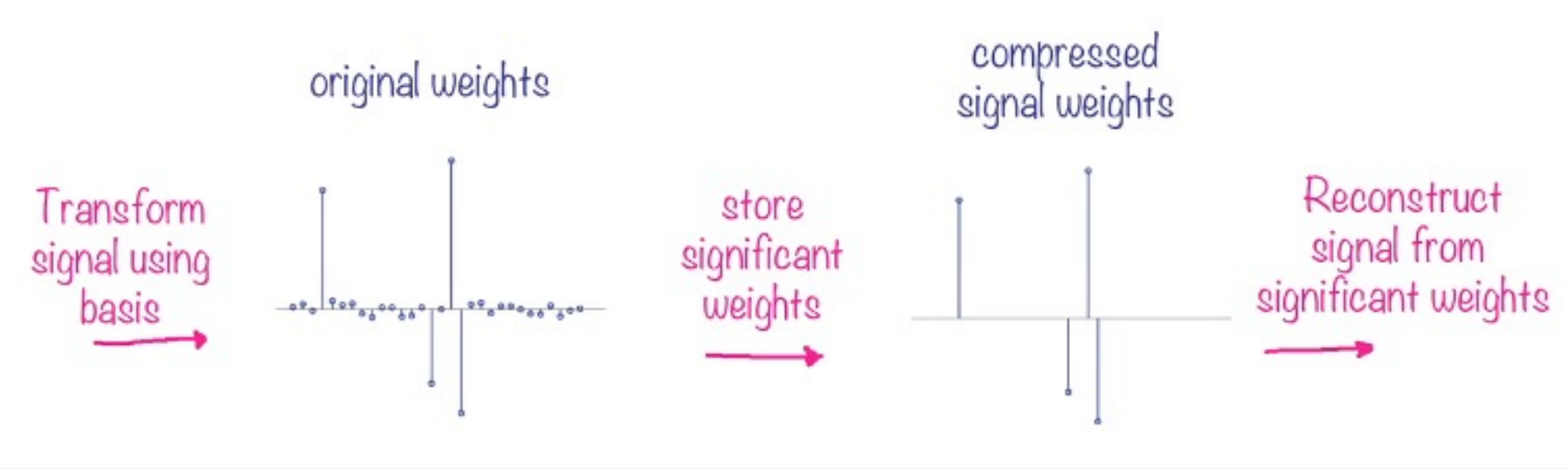
Using Basis Expansions: Filtering



Using Basis Expansions: Denoising



Using Basis Expansions: Compressing



Summary

- Basis expansions for signals have a myriad of uses in signal processing.
- The general idea is to express an arbitrary signal as a weighted sum (or integral) of basis signals.
- There are many different types of bases. Good bases concentrate the signal information into a small number of weights.
- The big picture view presented in this lesson will make it much easier to understand basis expansions and their role in signal processing.

Reference

- Prof. Barry Van Veen, 2014, Digital Signal Processing
- S. Salivahanan, Vallavaraj, (2001), Digital Signal Processing, McGraw-Hill, Singapore
- Mitra, Sanjit Kumar, (2001), Digital Signal Processing: A Computer-Based Approach, 2nd edition
- Heck, Bonnie S; Kamen, Edward W, (1997), Fundamentals of Signals and System UsingMatlab, Prentice Hall, New Jersey